BUS 9430 · CAPSTONE 2 · DATA UNDERSTANDING AND PREPARATION

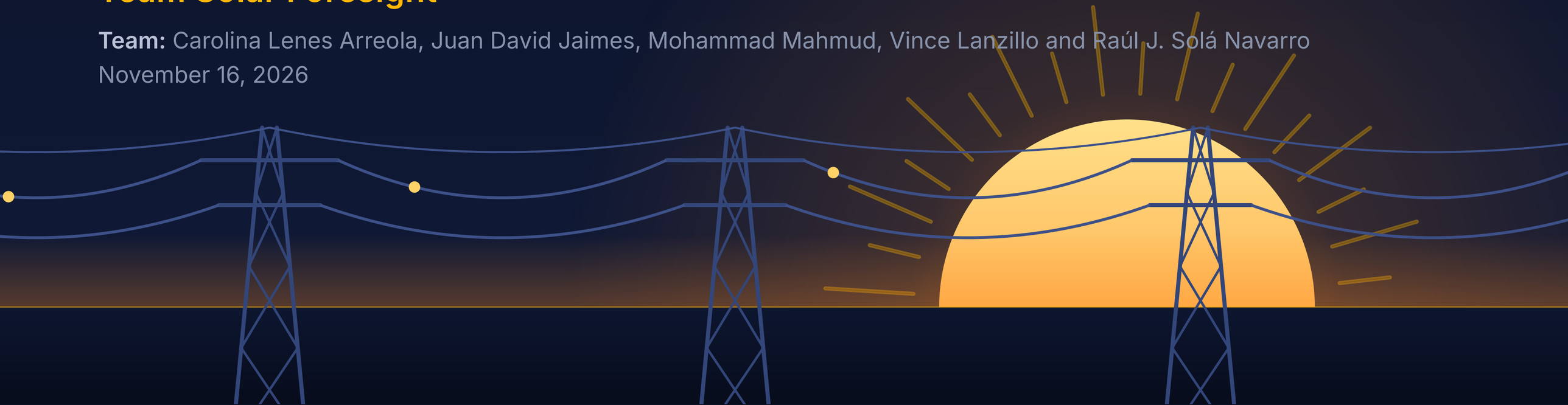
Tomorrow's Sun, Today's Plan

What our data shows, and how we get it ready

Team Solar Foresight

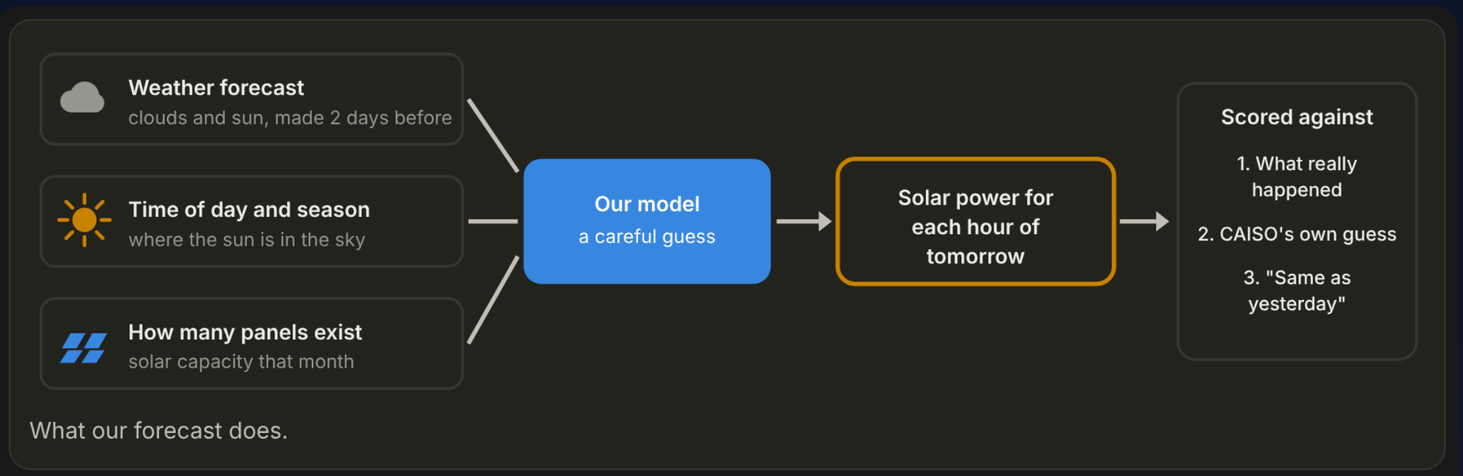
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November 16, 2026



THE BUSINESS PROBLEM

Guessing tomorrow’s solar power, hour by hour



What our forecast does.

Why it matters

- Fewer last-minute power buys.
- Less power wasted.
- Fewer backup gas plants running.

Who it helps

- Grid planners.
- Companies that buy power.
- Anyone who pays an electric bill.

How it is different

- Only free public data.
- A fair scorecard against CAISO and simple guesses.
- Every step shown, so anyone can check it.

TARGET VARIABLE

Solar power for each hour of tomorrow

In megawatts (MW), for the whole CAISO grid.

THE RULE

Only what we know at 10 a.m.

CAISO's bids close at 10 a.m. Pacific, the day before.

Picture from our plain-language primer (docs/solar-primer.html in the team repository).

WHAT WE FORECAST

We forecast solar output, not electricity demand

TOTAL DEMAND
What customers use

−

WIND AND SOLAR
What the weather supplies

=

NET LOAD
What gas, batteries and imports must cover

DEMAND

A crowded field with a smaller miss

CAISO, utilities and vendors all forecast it. It misses by about 2% to 3%. Solar misses by about 10% of its output.

OUR TARGET VARIABLE

Hourly solar output, in megawatts (MW)

How much power utility-scale solar plants in the CAISO area actually produce each hour, as measured by the U.S. Energy Information Administration.

Net load definition: U.S. Energy Information Administration (EIA).

OBTAIN AND QUALIFY THE DATA

Five public sources, one hourly table

ROLE	SOURCE	WHAT IT GIVES US	WHAT WE PULLED
Answer key	EIA Hourly Electric Grid Monitor	Actual hourly solar output for the CAISO area	Hourly, Jan 2024 to Oct 17, 2026
Inputs	Open-Meteo Previous Runs	Weather forecasts published before the 10 a.m. bid close	51 locations across California, Mar 2024 to Oct 2026
Benchmark	CAISO OASIS, report SLD_REN_FCST	CAISO's own day-ahead solar forecast	Hourly, same dates, three solar regions added up
Scale	EIA Form 860M	Installed solar capacity, so errors compare fairly over time	Monthly, Mar 2024 to Sep 2026
Flags	CAISO curtailment files	Solar that was turned down because the grid could not use it	Mar 2024 to Oct 2026

How we qualified them: public and free, documented, covering the same period (954 days), and usable at 10 a.m.

Weather data by Open-Meteo.com (CC BY 4.0). All data files are saved in the team repository.

THE RULE THAT KEEPS THE TEST HONEST

We may only use what is known at 10 a.m.

ALLOWED (MAIN MODEL)

Weather forecast made two days before

Published before the 10 a.m. bid close for every hour of the next day. This is the input we use.

COMPARISON ONLY

Weather forecast made one day before

For afternoon hours it is likely published after the bid close. We run it only as a labeled comparison, never as our result.

Why it matters: a forecast that peeks at later information looks better than it could be in real life.

Both versions are in the data, so the comparison is a single switch.

DATA UNDERSTANDING

The data at a glance: nearly complete

PERIOD COVERED

Mar 2024 to Oct 2026

954 days, from March 8, 2024 to October 17, 2026.

HOURS OF DATA

22,896

One row per hour. 11,733 of them are daylight hours, when solar can make power.

HOURS WITH A GAP

< 1%

226 hours across all sources, out of 22,896. Gaps are in the weather feed (129), the CAISO forecast (72) and EIA (25).

So the data is in good shape before we change anything.

PREPARE THE DATA

From five files to one analysis-ready table

1

Profile

Look at every file

22,896 hourly rows. 226 hours with a gap, under 1%.

2

Organize

Put it on one clock

EIA's hour labels ran 1 hour late until Dec 2, 2025. We corrected them.

3

Join

One row per hour

Five files joined on the hour. Rows counted after every join: always 22,896.

4

Enrich

Add what the model needs

Sun up or not, season, installed capacity that month, a turned-down flag.

5

Segment

Split by time

Earliest days to train and tune. The latest stay locked for the final test.

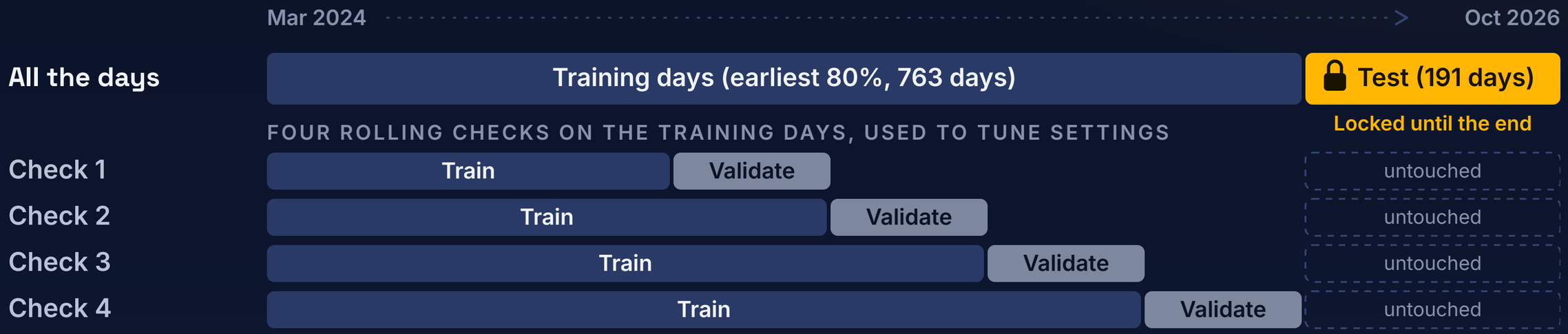
Mar 2024 — Oct 2026

Training and tuning: earliest 763 days

Locked test: 191 days

DATA PREPARATION

Test on days the model has never seen



1

Train

The earliest 80% of days (763 days, to April 9, 2026) teach the models.

2

Validate

Four rolling checks of about three months each compare the models and tune them.

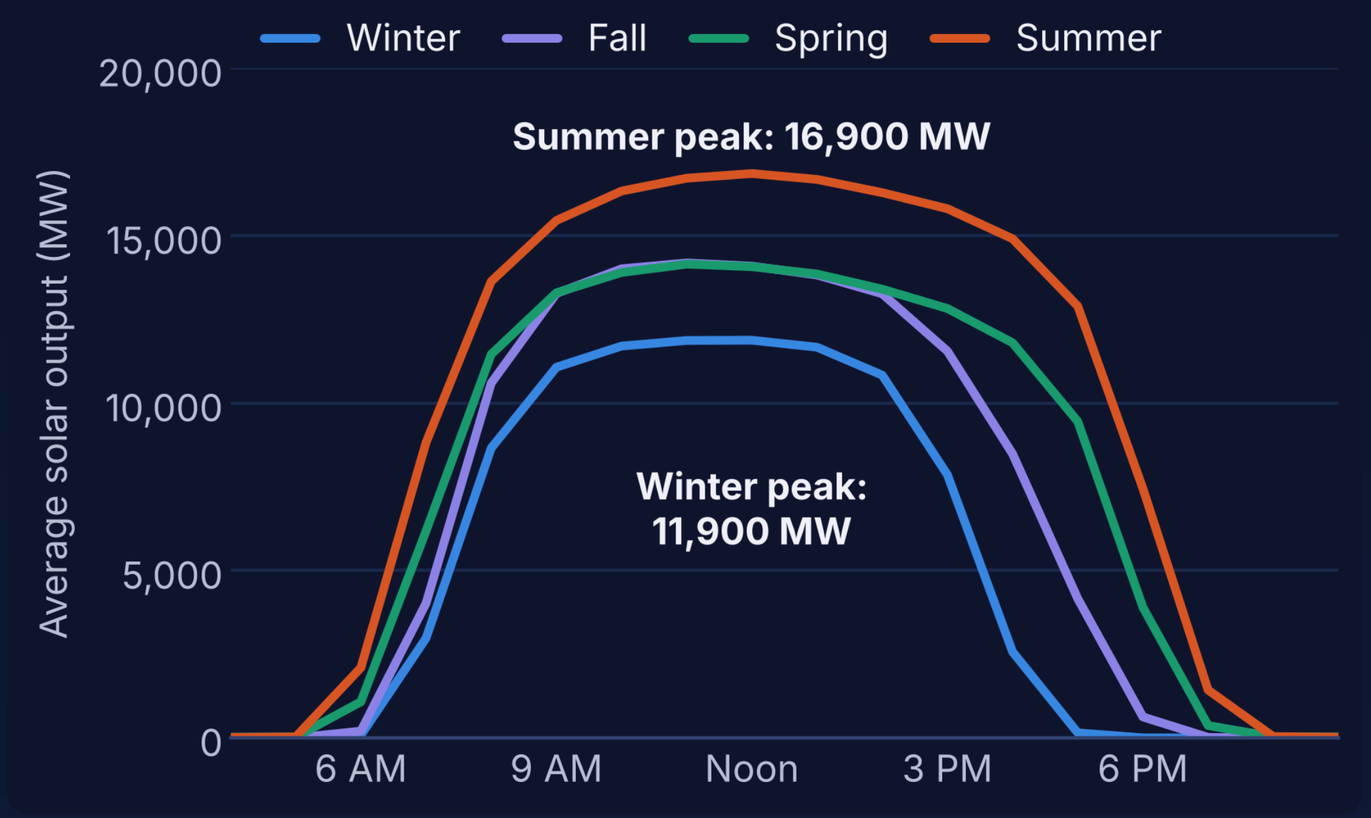
3

Test

The latest 191 days (April 10 to October 17, 2026), scored once at the end. No winter, so winter is judged in validation.

DESCRIPTIVE ANALYTICS

A typical solar day: up at 6, peak at noon, done by 8



+42%

Summer's midday peak against winter's: about 16,900 MW versus 11,900 MW.

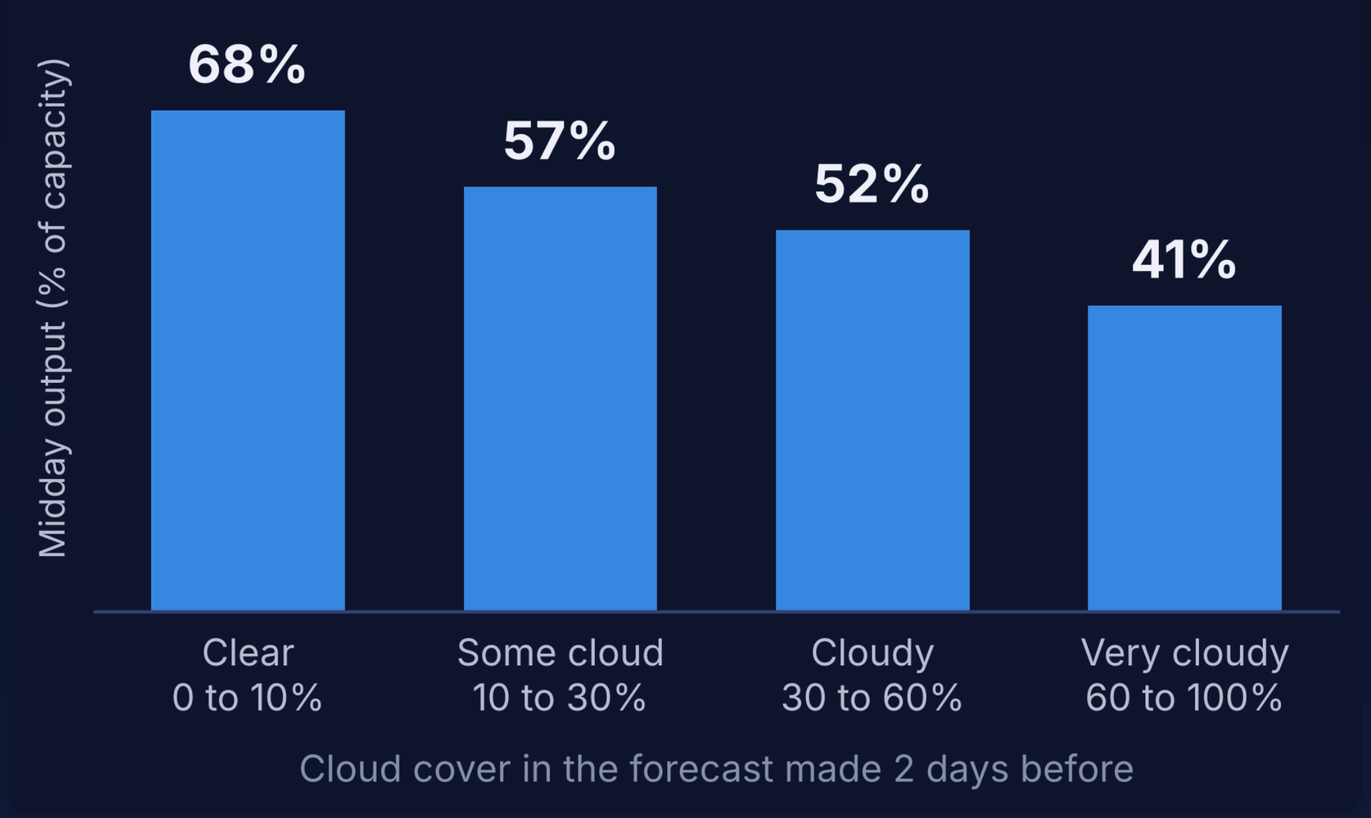
2 inputs

Hour of day and season. The model has to know both, so we give them to it directly.

Chart made in Python from our hourly table. Average over training and validation days, Mar 2024 to Apr 9, 2026. Winter is December to February.
Source: EIA.

DESCRIPTIVE ANALYTICS

Cloudier forecast, less solar



0.88

How closely forecast sunshine tracks actual output, where 1.0 is perfect. Measured across 9,190 daylight hours.

59%

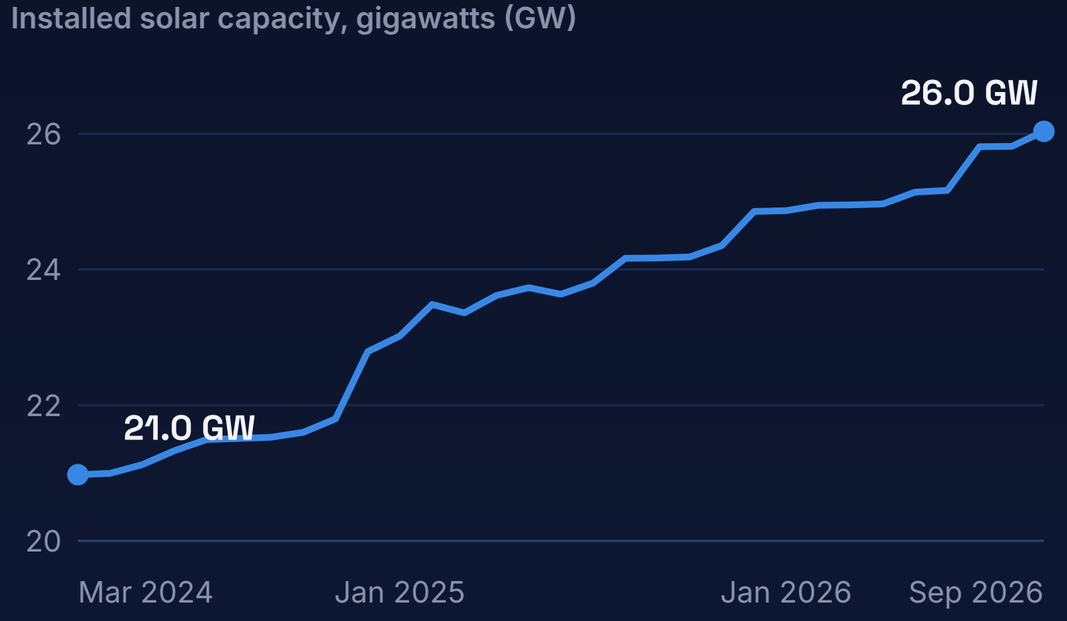
Biggest one-day drop: on March 31, 2026, output fell from 133 to 55 gigawatt-hours (GWh) in a day.

Chart made in Python from our hourly table. Midday = 10 a.m. to 1 p.m. Pacific. Cloud cover is the forecast made two days earlier. All seasons, so winter's lower sun also plays a part. Sources: EIA, Open-Meteo.

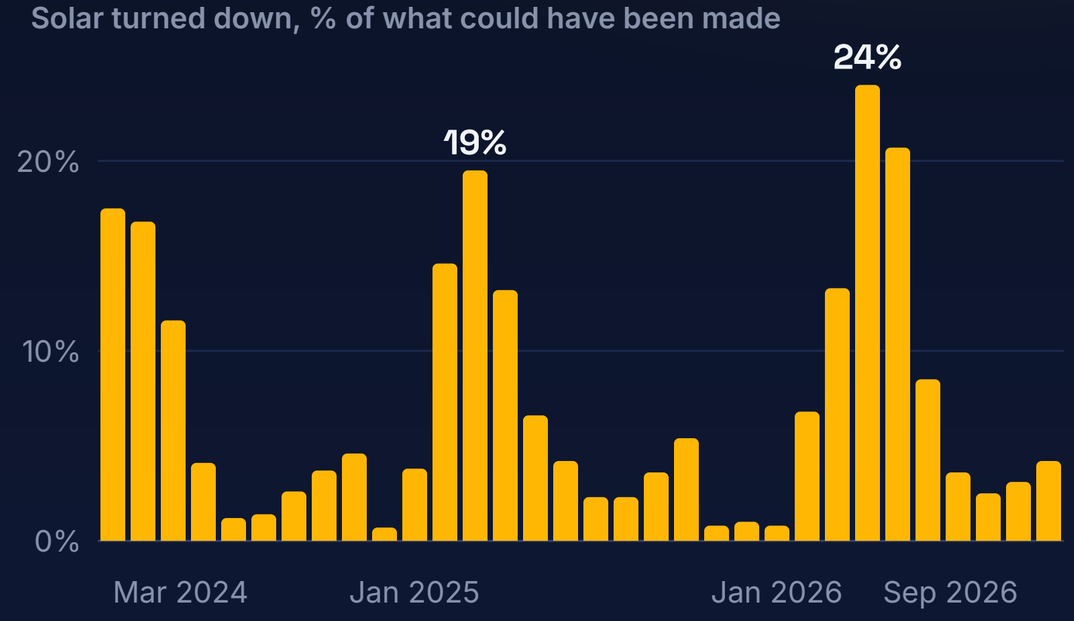
DATA UNDERSTANDING

More panels every month, and more solar turned down

Installed solar capacity grew 24%



Spring is when solar gets turned down



WHY IT MATTERS

Errors must be scaled by capacity

A 1,000 MW miss meant more in 2024 than it will in 2026. We report errors as a share of installed capacity.

WHY IT MATTERS

Turned-down solar hides true output

In April 2026 about 1 in 4 possible megawatt-hours was turned down. Over the whole period it is 7.5%.

Sources: EIA Form 860M (capacity, includes about 2.8 GW in Arizona and Nevada); CAISO curtailment reports.

DATA QUALITY

How we check quality, and what we did

WHAT WE FOUND	HOW WE FOUND IT	WHAT WE DID
Clocks off by 1 hour	Compared EIA's daily curve with CAISO's own numbers and the sun's position.	Moved EIA's labels back 1 hour before Dec 2, 2025. They now match CAISO (correlation 0.998).
Missing hours	Counted the blanks in every file.	226 of 22,896 hours: weather 129, CAISO forecast 72, EIA 25. Flagged and left out of the scores.
Odd values at night	Checked output while the sun was below the horizon.	Small negatives (down to 74 MW below zero) set to 0 in a working copy, original kept. 351 odd hours flagged and left out of training.
Turned-down solar	Compared CAISO's curtailment reports with EIA output.	7.5% of possible solar, up to 24% in April 2026. Flagged, so we can report with and without.
More panels over time	Compared monthly capacity files, Mar 2024 to Sep 2026.	Capacity is up 24%, so errors are scaled by installed capacity.

Raw files are never edited. Every fix is code that anyone can re-run.

PRIVACY AND DELIVERY RISKS

Privacy is not a concern. That is the point.

PRIVACY

Open data, nothing personal

Public, grid-wide numbers and weather. No customers, no homes, no personal information.

So the whole model can be open: a base, or as is, for anyone's business case.

DELIVERY RISK 1

Free weather data has limits

Non-commercial license and a daily cap (we used about 6,500 of 10,000 calls). A business needs the paid tier.

DELIVERY RISK 2

Sources can change without notice

EIA changed its clock labels on Dec 2, 2025. Our pull script and timing check can be re-run.

DELIVERY RISK 3

Short history

2.6 years: each month appears 2 or 3 times, and our test period has no winter.

Next, in Capstone 3: build and compare forecasting models on this table.

QUESTIONS

Questions, and your input

YOUR INPUT

Is there anything in the data or the test design you would want us to check?

We would rather hear it now than after we build models.

WHAT COMES NEXT

Models, built on this table

The table is analysis-ready: one row per hour, flags for every gap and every turned-down hour, and a locked test set.

The slides after this one are backup answers to questions we expect.

Q&A BACKUP

If the guess is too low, is the extra solar lost?

1

Sold

Real-time market

The farm sells what it made beyond its plan, at the real-time price. At noon that can be near $0 or below.

2

Others back down

Every 5 minutes

CAISO sends each plant a new target. Gas plants and dams make a bit less.

3

Batteries fill

Peak charging at noon

Batteries in CAISO's area grew from 8.0 GW (2023) to 11.6 GW (2024).

4

Sent to neighbors

Western energy market

Avoided 274 GWh of turn-downs in 2024, about 8% of the total.

5

Turned down

Only at the end

By price signal or, rarely, a direct order. Inverters make the panels deliver less.

SO WHAT IS "LOST"?

Nothing is dumped. The power is never made.

The sunlight is not converted, so there is nothing to throw away or make safe. What is lost is clean energy and the owner's sales.

Sources: EIA Today in Energy (curtailment in California, 2024); GridStatus (spring 2026); CAISO market monitoring 2025 annual report; arXiv 2011.09555. Step 1 follows the standard two-settlement rule, not checked line by line in CAISO's tariff.

Q&A BACKUP

How the grid is protected from too much power

If more power goes in than comes out, grid frequency rises above 60 Hz and can lead to instability. Too little power is the more dangerous side.

LAYER 1, SLOWEST

The day-ahead plan

Sets most of tomorrow's mix. This is the layer our forecast helps.

LAYER 2

The market, every 5 and 15 minutes

Plants get new targets as real use and real sun come in.

LAYER 3

Regulation, every 4 seconds

A control computer sends small up and down signals. Full response within about 10 minutes.

LAYER 4

Governors, within seconds

Plants react on their own to a frequency change as small as 36 mHz.

LAYER 5

Turn-down orders

Solar farms are told to make less, as on the previous slide.

LAYER 6, LAST RESORT

Safety relays

Disconnect equipment if frequency leaves safe limits. General practice, settings not checked.

Sources: CAISO frequency response issue paper (layers 3 and 4, and the frequency statement). Rooftop solar cannot be commanded one system at a time and shows up as lower demand.

Q&A BACKUP

Why solar, and not demand?

	DEMAND (TOTAL USE)	SOLAR
Typical miss in MW	about 650 to 750 MW	about 1,100 MW in a daylight hour
Miss as a share of what is forecast	2.3% to 3.4% of demand	about 10% of solar output (1.1 GW out of about 11 GW)
Fastest move in 2 hours	demand up 1.9 to 2.7 GW	solar down 7.2 to 9.2 GW
What drives it	temperature, habits, holidays	clouds, over the whole state at once

- **Demand is not solved.** One outside test week (July 2024, a vendor study) found CAISO's load forecast 4.55% off on average.
- **Solar is the faster, bigger miss for its size,** and the grid really runs on demand minus solar.
- **A crowded field versus a free input.** CAISO, utilities and vendors forecast demand. Free weather data drives solar directly.

Demand column: 4 sample days (15th of Jan, Apr, Jul, Oct 2025), CAISO daily files. Solar column: CAISO as published, 4.7% of 23.6 GW average capacity. Our table already holds temperature, so a demand model could come later.

Q&A BACKUP

What a solar miss is worth, in watts and dollars

EACH POINT OF MISS

1.1 TWh

a year, about 240 MW in a daylight hour, about $43 million at $41 per MWh.

CAISO'S MISS, 4.7%

5.0 TWh

1.8% of all California power, about $204 million at $41 per MWh.

THE REAL COST IS LOWER

$205M

CAISO's real-time imbalance offsets in 2025. A solar miss is one cause among several.

WHAT AN OPEN MODEL COULD BE WORTH

About 1.5 TWh a year, roughly $61 million at the average price

That is the gap between copying 2 days ago (6.1% miss) and CAISO's level (4.7%), for anyone without CAISO's forecast. A size, not a promise.

Method: miss % x 23.6 GW average solar capacity x 4,489 daylight hours a year. $41 per MWh is CAISO's 2024 average day-ahead price. California uses 273 TWh a year (31 GW on average). Sources: California Energy Commission; CAISO market monitoring 2024 and 2025 annual reports.

APPENDIX A

Data dictionary: the main columns

COLUMN	MEANING	UNIT	SOURCE
`hour_end_utc`	The hour ending at this time, UTC (the key for joining all sources)	timestamp	All, aligned
`eia_mw`	Actual utility-scale solar output (target variable)	MW	EIA
`caiso_dam_mw`	CAISO day-ahead solar forecast (benchmark only)	MW	CAISO OASIS
`capacity_mw`	Installed solar capacity in that month	MW	EIA Form 860M
`cloud_cover`, `shortwave_radiation`, `temperature_2m`, …	Weather forecast inputs, two-day-ahead version, capacity-weighted across locations	%, W/m², °C	Open-Meteo
`curt_mwh`	Solar energy turned down in that hour	MWh	CAISO curtailment files

Full list of columns, units and checks: docs/DATA_DICTIONARY.md in the team repository.

APPENDIX B

Sources

U.S. EIA, Hourly Electric Grid Monitor and API; net load definition: eia.gov/opendata

U.S. EIA, Form 860M, monthly installed capacity: eia.gov/electricity/data/eia860m

Open-Meteo, Previous Runs API, CC BY 4.0: open-meteo.com/en/docs/previous-runs-api

CAISO, day-ahead market (bids close 10 a.m. Pacific, results about 1 p.m.): caiso.com/market-operations/products-services

CAISO OASIS, day-ahead solar forecast, report SLD_REN_FCST: oasis.caiso.com

CAISO, production and curtailments data (curtailment workbooks and daily renewables report): caiso.com/library/production-curtailments-data

CAISO Department of Market Monitoring, 2024 and 2025 annual reports on market issues and performance: caiso.com/documents/2025-annual-report-on-market-issues-and-performance.pdf

U.S. EIA, curtailment in California: eia.gov/todayinenergy/detail.php?id=65364

CAISO, frequency response issue paper: caiso.com/documents/issuepaper_frequencyresponsephase2.pdf

Team repository, raw data, data dictionary, FAQ and how-to: github.com/RaulSolaNavarro/caiso-solar-forecast